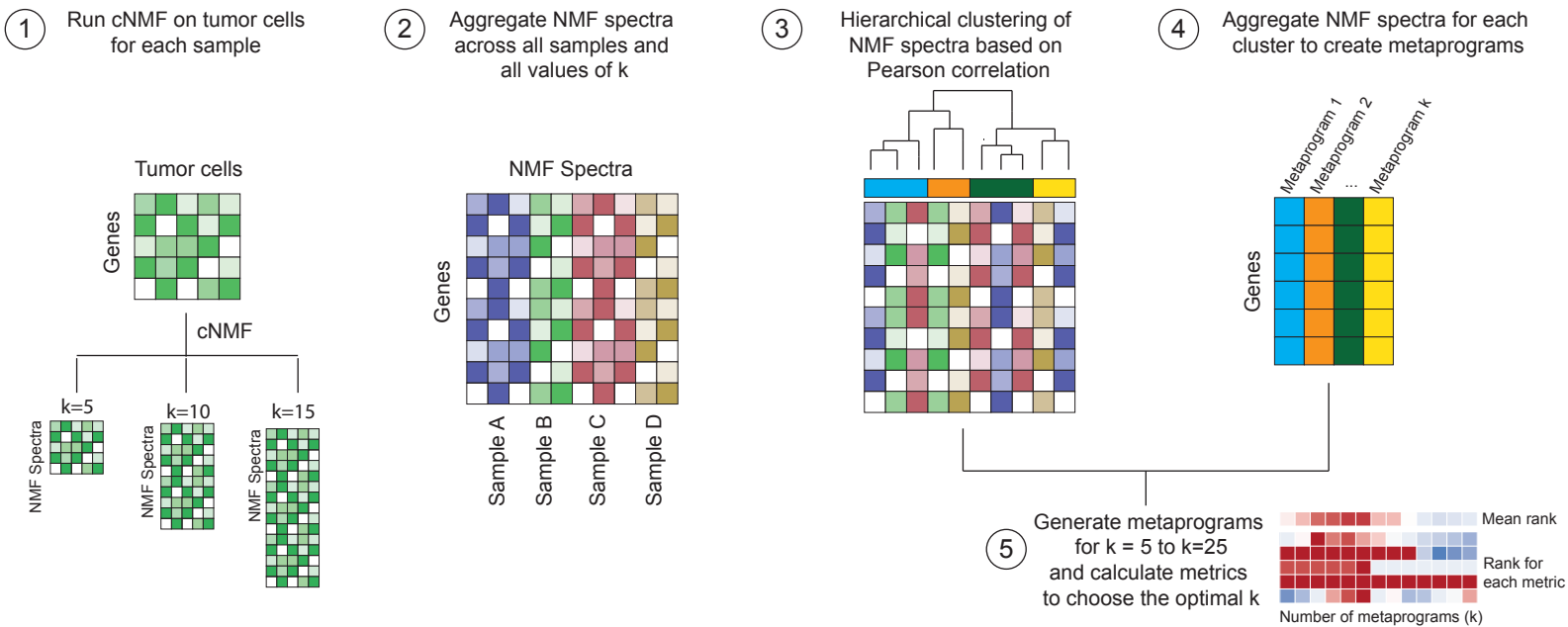

1
Run cNMF on tumor cells for each sample
Tumor cells
Genes
cNMF
k=5
k=10
k=15
NMF Spectra
NMF Spectra
NMF Spectra
2
Aggregate NMF spectra across all samples and all values of k
NMF Spectra
Genes
Sample A
Sample B
Sample C
Sample D
3
Hierarchical clustering of NMF spectra based on Pearson correlation
4
Aggregate NMF spectra for each cluster to create metaprograms
Metaprogram 1
Metaprogram 2
...
Metaprogram k
Genes
5
Generate metaprograms for k = 5 to k=25 and calculate metrics to choose the optimal k
Mean rank
Rank for each metric
Number of metaprograms (k)